

Linear Algebra and its Applications

Assignment 6

Due: 31 March 2026

Instructions: The solutions to this assignment are to be uploaded on Moodle as a PDF file. You may prepare a Jupyter Notebook and save it as a PDF output.

1. Consider the matrix

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

- (a) Implement the power method in Python (use the in-built matrix-vector multiplication only).
- (b) Starting from two different random (and linearly independent) initial vectors, plot the iterates $x^{(k)}$ on the unit circle.
- (c) Compute the true dominant eigenvector using a library routine and plot it on the same figure.
- (d) Describe the convergence behavior you observe.

2. Consider

$$A_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1.9 \end{bmatrix}.$$

- (a) Apply the power method to both matrices.
- (b) Plot the error $\|x_k - v_1\|$ vs iteration number.
- (c) Compare convergence rates and try to find if there is any relationship with the difference between eigenvalues. Describe your findings.

3. Consider

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}.$$

- (a) Implement inverse iteration to compute the smallest eigenvalue.
- (b) Implement shifted inverse iteration with shifts: $\mu = 3$, $\mu = 3.5$
- (c) For each case, record the number of iterations needed for convergence.
- (d) Explain how the choice of shift affects convergence.

4. Using the same matrix as above:

- (a) Implement Rayleigh Quotient Iteration.
- (b) Track the error: $|\lambda_k - \lambda|$ at each iteration.
- (c) Plot the error on a logarithmic scale.
- (d) Compare convergence with power method and shifted inverse iteration.
- (e) What do you observe about the speed of convergence?

5. Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

- (a) Run Rayleigh Quotient Iteration for different initial vectors.
- (b) Observe which eigenvalue the algorithm converges to.
- (c) Repeat the experiment for shifted inverse iteration.
- (d) Explain how the initial vector influences the result.